

Bias-Variance Trade-off

Measuring Bias and Variance

- Create B bootstrap variants of the dataset
- Find out the predictions for each of the B bootstrap variants
- Compute the average of all the predictions $\overline{S^*} = \frac{1}{B} \sum_{b=1}^B S(\mathbf{z}^{*b})$
- Bias: average across all models of the difference between the model prediction and the average prediction

$$Bias^2 = \sum_{i=1}^n (S(z_i^{*b}) - \overline{S^*})^2$$
$$\widehat{Var}[S(Z)] = \frac{1}{B-1} \sum_{b=1}^B (S(\mathbf{z}^{*b}) - \overline{S^*})^2$$

Bias-Variance Decomposition of Squared Error

- Assume that $y = f(\mathbf{x}) + \epsilon$
 - Noise ϵ is sampled from a normal distribution with 0 mean and variance σ^2 :
 $\epsilon \sim \mathcal{N}(0, \sigma^2)$
 - Noise lower-bounds the performance (error) we can achieve

- Recall the objective function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n \left(y^{(i)} - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) \right)^2$$

- We can represent the function as an approximation of the expected value of the squared error: $E(y - h_{\boldsymbol{\theta}}(\mathbf{x}))^2$

Bias-Variance Decomposition of Squared Error

$$y = f(x) + \epsilon$$

$$E(y - h_{\theta}(x))^2 = E \left[\underbrace{(y - f(x))}_a + \underbrace{(f(x) - h_{\theta}(x))}_b \right]^2$$

$$\begin{aligned} & E \left[(y - f(x))^2 + 2(y - f(x))(f(x) - h_{\theta}(x)) + (f(x) - h_{\theta}(x))^2 \right] \\ &= E \left[(y - f(x))^2 \right] + 2E \left[(y - f(x))(f(x) - h_{\theta}(x)) \right] + E \left[(f(x) - h_{\theta}(x))^2 \right] \\ &= E \left[\epsilon^2 \right] + \underbrace{2E \left[\epsilon (f(x) - h_{\theta}(x)) \right]}_{=0} + E \left[(f(x) - h_{\theta}(x))^2 \right] \end{aligned}$$

First term: $E \left[\epsilon^2 \right] = \text{Var}(\epsilon) + (E[\epsilon])^2 = \sigma^2$

Second term: $2E \left[\epsilon (f(x) - h_{\theta}(x)) \right] = 2E[\epsilon] E[f(x) - h_{\theta}(x)] = 0$
(Noise is uncorrelated)

3rd term: $E[(f(x) - g(x))^2]$

$$= E\left[\left(f(x) - \bar{h}(x) + (\bar{h}(x) - h(x))\right)^2\right]$$

$$\boxed{\bar{h}(x) = E[h(x)]}$$

$$= E\left[(f(x) - \bar{h}(x))^2 + 2(f(x) - \bar{h}(x))(\bar{h}(x) - h(x)) + (\bar{h}(x) - h(x))^2\right]$$

$$= E\left[(f(x) - \bar{h}(x))^2\right] + 2E\left[(f(x) - \bar{h}(x))(\bar{h}(x) - h(x))\right] + E\left[(\bar{h}(x) - h(x))^2\right]$$

$$E[\bar{h}(x) - h(x)] = 0 \Rightarrow \text{Cross term} = 0$$

$$\rightarrow \text{Bias}^2(h(x)) + E\left[(h(x) - E[h(x)])^2\right]$$

$$\therefore \text{Bias}^2(h(x)) + \text{Var}(h(x))$$

$$E[(y - h_\theta(x))^2] = \sigma^2 + \text{Bias}^2(h(x)) + \text{Var}(h(x))$$

Regularization

- Linear regression objective function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2} \sum_{j=1}^d \theta_j^2$$

λ is the regularization parameter.

Bias-Variance Illustration

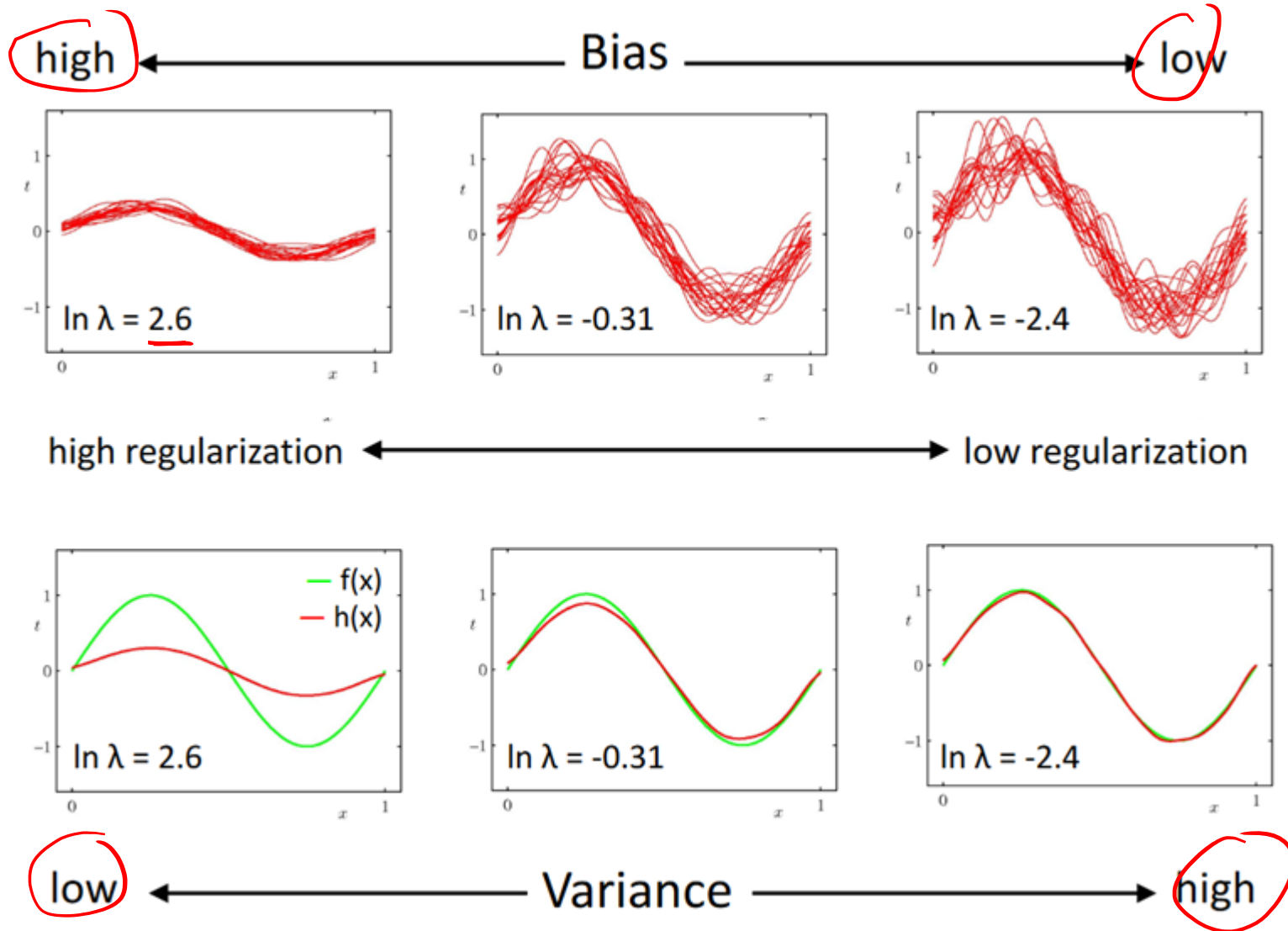


Illustration of dependence of bias and variance on model complexity (sinusoidal dataset): There are 100 datasets, each having 25 datapoints, there are 24 Gaussian basis functions in the model.

Top row: Result of fitting the model to the dataset for various values of $\ln \lambda$ (20 out of 100 fits are shown)

Bottom Row: Corresponding average of the 100 fits (red) along with the sinusoidal functions from which the dataset is generated (green).

Bias-Variance Illustration

- Reducing training error drives down bias but ignores variance

